

# Cedar Site Bai

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## ★ OBJECTIVE

Ph.D. cand. in CS with a strong background in machine learning, optimization, and statistics. Experienced in their applications to language model pre-training, fine-tuning, recommendation, and AI safety, with **oral presentations at ICML, ICLR, COLT and internships at Amazon, TikTok**. Seeking a **Researcher/Scientist** role.

## 🎓 EDUCATION

- **Purdue University** Jan. 2021 – May. 2026 (Exp.)  
*Ph.D. in Computer Science (Machine Learning & Optimization)* Advisor: [Prof. Brian Bullins](#)  
*M.S. in Statistics & Computer Science (along the way)* West Lafayette, IN, US
- **Xi'an Jiaotong University (XJTU)** Sep. 2016 – Jun. 2020  
*B.S. in Computer Science (Qian Xuesen Honors College)* Xi'an, China  
Selected Awards: "Siyuan" Scholarship (twice); National 2<sup>nd</sup> Prize in Mathematical Contest in Modeling (CUMCM, top 3%)

## 🧰 INDUSTRY EXPERIENCE

- **Amazon**, Prime Video Personalization, Applied Scientist Intern May. 2025 – Sep. 2025  
Work on **conversational recommendation** Mentors: Drs. [Duanshun Li](#), [Zhenyu Liao](#)
  - Designed a reward quantifying information gain via uncertainty reduction in multi-turn conversations
  - Built a data-generation pipeline integrating LLM recommenders, judges, and user simulators using real-world data
  - Fine-tuned LLMs with SFT, DPO, PPO, increasing conversation efficiency by 7% and hit rate by 2%
- **TikTok**, Shop Mall Ads, Machine Learning Engineer Intern May. 2024 – Aug. 2024  
Work on **ads recommendation ranking models** Mentor: [Du Zhang](#), Manager: [Dr. Yan Yan](#)
  - Delivered two launches that improved advertiser value by 1.63% and 5.94% (**\$X00,000 in profits**) after A/B test
  - Applied neural architecture search to select features/emb. sizes, reducing the model size & improving accuracy
  - Adapted Deep Cross Network for improved conversion rate (CVR) prediction on sequential data

## 📖 SELECTED PUBLICATION

- [1] **[ICML'25, Oral (Top 1%)] C. S. Bai\***, A. Y. Zheng\*, B. Bullins, R. A. Yeh. Model Immunization from a Condition Number Perspective. [\[PDF\]](#)
- [2] **[ICML'25] C. S. Bai\***, X. Luo\*, B. Li\*, P. Drineas, R. Zhang, B. Bullins. Stacey: Promoting Stochastic Steepest Descent via Accelerated  $\ell_p$ -Smooth Nonconvex Optimization. [\[PDF\]](#)
- [3] **[COLT'25] C. S. Bai**, B. Bullins. Faster Acceleration for Steepest Descent. [\[PDF\]](#)
- [4] **[ICLR'25, Oral (Top 1.7%)] C. S. Bai**, B. Bullins. Tight Lower Bounds under Asymmetric High-Order Hölder Smoothness and Uniform Convexity. [\[PDF\]](#)
- [5] **[IJCAI'21] H. Zhang, S. Bai**, X. Lan, D. Hsu, N. Zheng. Hindsight Trust Region Policy Optimization. [\[PDF\]](#)

## 🔧 PROFESSIONAL TECHNIQUES

**Machine learning (ML):** Optimizer design/analysis, reinforcement learning/RLHF, recommendation, unlearning;  
**Programming:** Python, MATLAB (**Proficient**); Shell script, C/C++, R, Java, SQL, HTML;  
**Packages / Tools:** PyTorch, NumPy, Linux Command, Git (**Proficient**); AWS, scikit-learn, TensorFlow, Pandas, Slurm.

## ADDITIONAL PUBLICATION

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- [1] **[Preprint]** C. S. Bai, A. Y. Zheng, R. A. Yeh, B. Bullins. Spectral Saliency for Machine Unlearning.
- [2] **[Preprint]** Z. Song, C. S. Bai, Z. Zhang, B. Bullins, D. Gleich. Decoupling Variance and Scale-Invariant Updates in Adaptive Gradient Descent for Unified Vector and Matrix Optimization.
- [3] **[Preprint]** H. Zhang, T. Hu, C. S. Bai, A. Xiao, D. Hsu. Uncertainty Estimation for Visual Grounding in the Open World.
- [4] **[ICLR'24]** S. Bai, B. Bullins. Local Composite Saddle Point Optimization. [\[PDF\]](#)
- [5] **[TMLR'24]** S. Bai, C. Ke, J. Honorio. On the Dual Problem of Convexified Convolutional Neural Networks. [\[PDF\]](#)
- [6] **[IROS'19]** H. Zhang, X. Lan, S. Bai, X. Zhou, Z. Tian, N. Zheng. ROI-based Robotic Grasp Detection for Object Overlapping Scenes. [\[PDF\]](#)
- [7] **[IROS'19]** H. Zhang, X. Lan, S. Bai, L. Wan, C. Yang, N. Zheng. A Multi-task Convolutional Neural Network for Autonomous Robotic Grasping in Object Stacking Scenes. [\[PDF\]](#)

## SELECTED COURSES

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**Machine Learning (ML):** Statistical ML, ML Theory, Computer Vision, Data Mining, Artificial Intelligence

**Math & Stats:** Optimization, Probability, Statistical Inference, Computational Stats, Multivariate Analysis

**Computer Science:** Data Structures, Algorithms, Database, Machine Structure, Operating Systems, Networks

## TEACHING & ACADEMIC SERVICE

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|---------------------------------------------------------------------|----------------------------------|
| Purdue CS571 Artificial Intelligence                                | TA (Fall 23)                     |
| Purdue CS182 Foundations of Computer Science (Discrete Mathematics) | TA (Spring 23)                   |
| Purdue CS251 Data Structures and Algorithms                         | TA (Fall 21, Spring 22, Fall 22) |

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| • ICLR 2026, 2025, 2024    | Reviewer |
| • NeurIPS 2025, 2024, 2023 | Reviewer |
| • ICML 2026, 2025, 2024    | Reviewer |
| • AISTATS 2025, 2023       | Reviewer |
| • AAAI 2026, 2025          | Reviewer |
| • TPAMI                    | Reviewer |
| • TMLR                     | Reviewer |